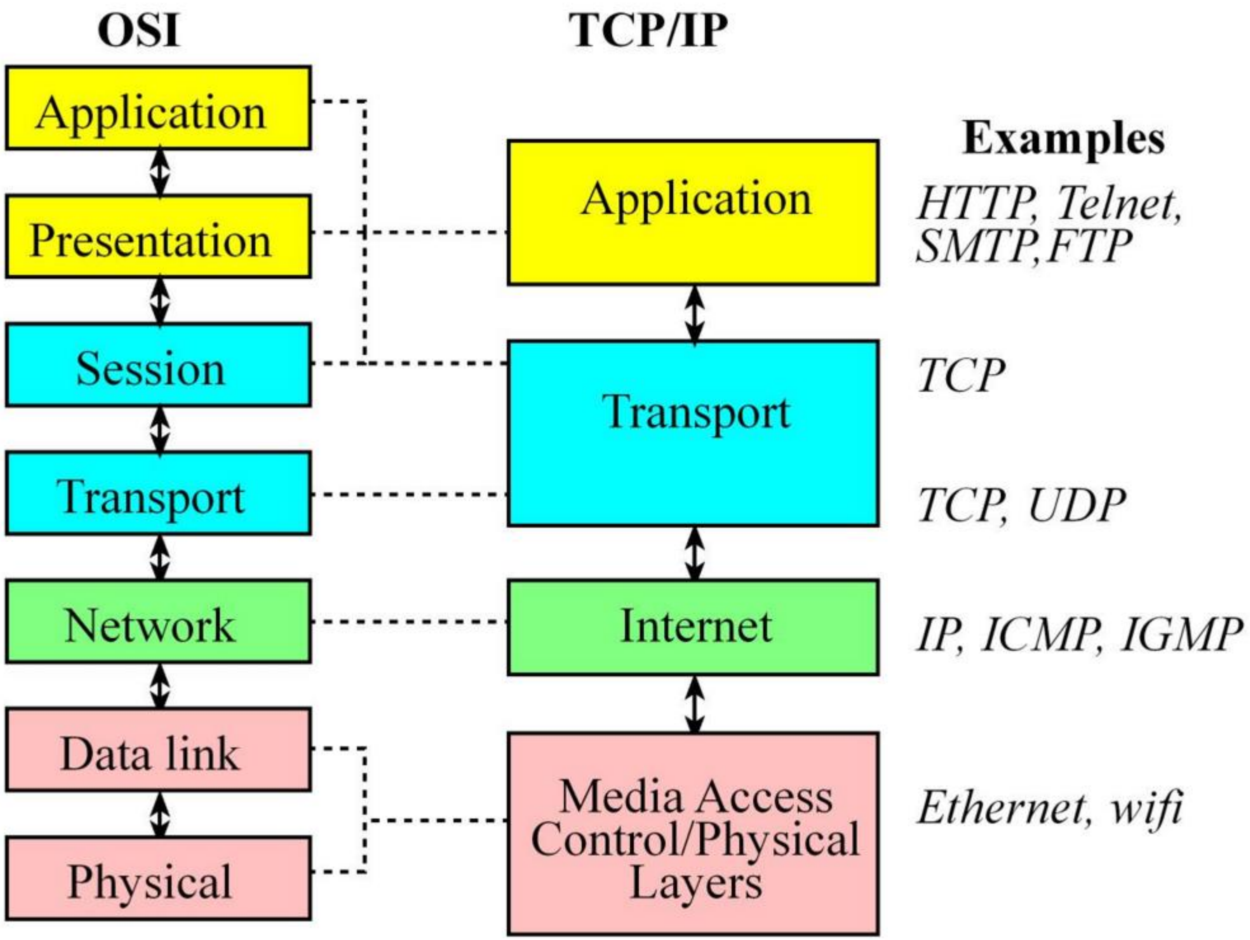
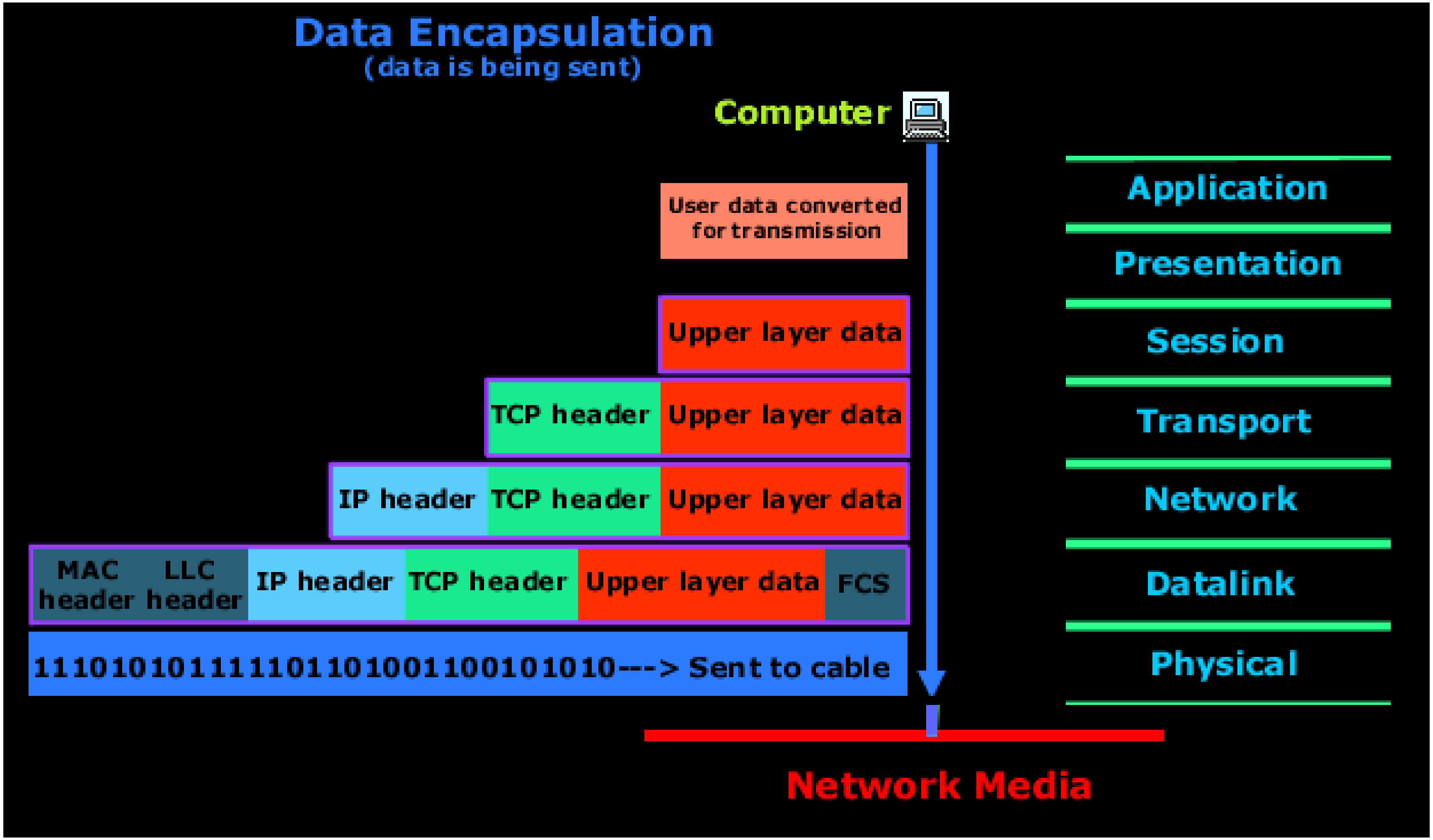
Socket Programming

OSI vs TCP/IP


OSI
TCP/IP
Application
Presentation
Session
Transport
Network
Data link
Physical
Examples
Application
HTTP, Telnet, SMTP,FTP
Transport
TCP
TCP, UDP
Internet
IP, ICMP, IGMP
Media Access Control/Physical Layers
Ethernet, wifi

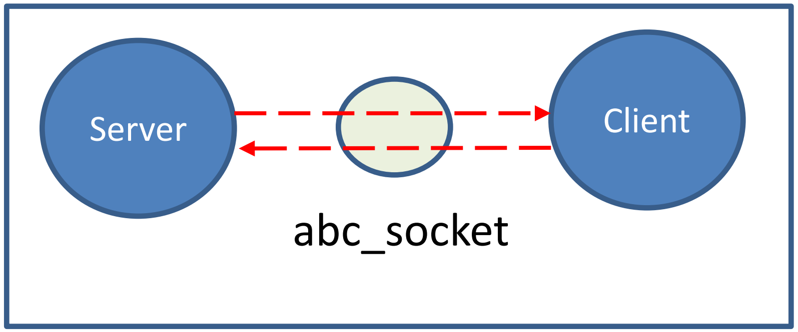
Data Encapsulation TCP/IP Model



Layer	Name	Data	Function
7	Application	Message	Meaning of data
6	Presentation	Message	Building blocks of data and encryption
5	Session	Message	Opening and closing of specific communication paths
4	Transport	Segment	Error checking
3	Network	Packet	Determination of data paths within the network
2	Data Link	Frame	Data transmission, source, destination, and checksum
1	Physical	Bitz	Voltage levels, signal connections, wire, or fiber

Socket (Communication End Point)

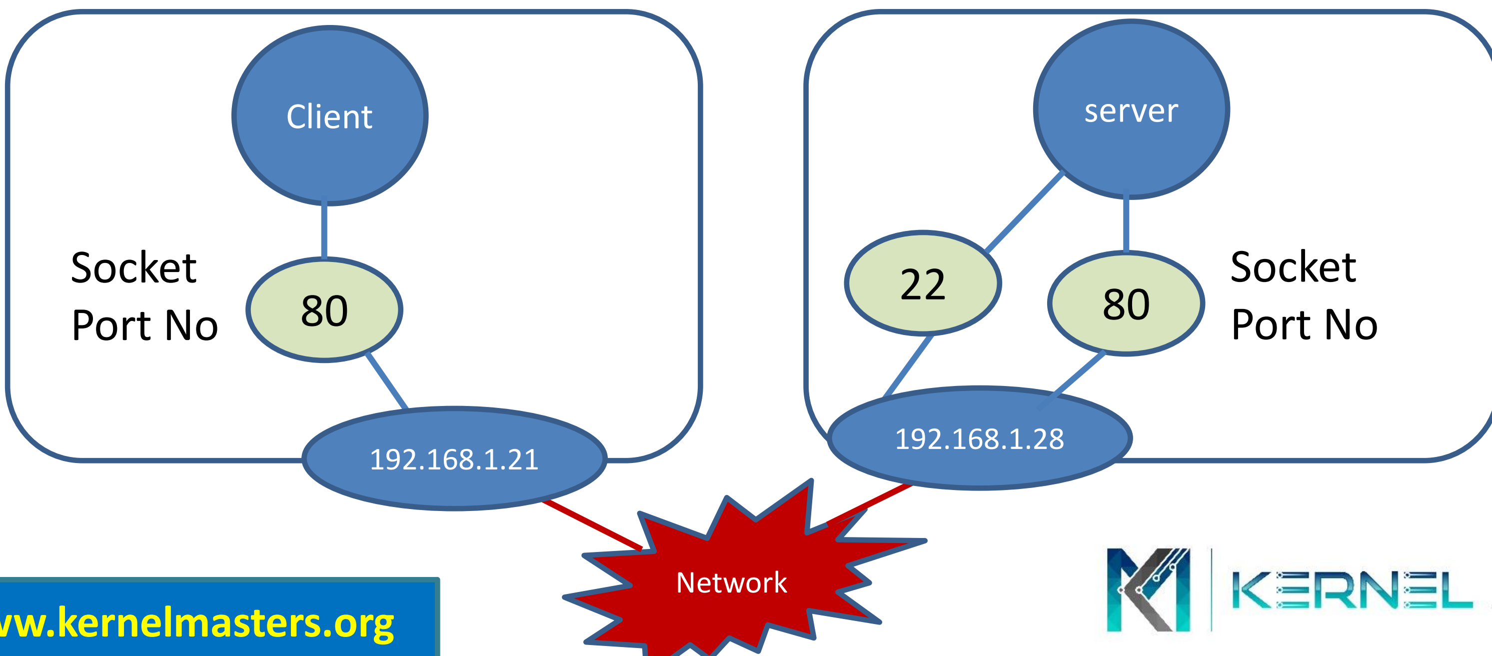
Named Socket: [With in the system]



https://www.iana.org/assignments/service-names-port-numbers/service-names-port-numbers.xhtml

Unnamed Socket:[comm. b/w system to system using Port No]

Client System **Server System**



Socket (Communication End Point)

office

Reception	Admin	Finance
Lab1	R&D1	HR
Lab2	R&D	R&D2

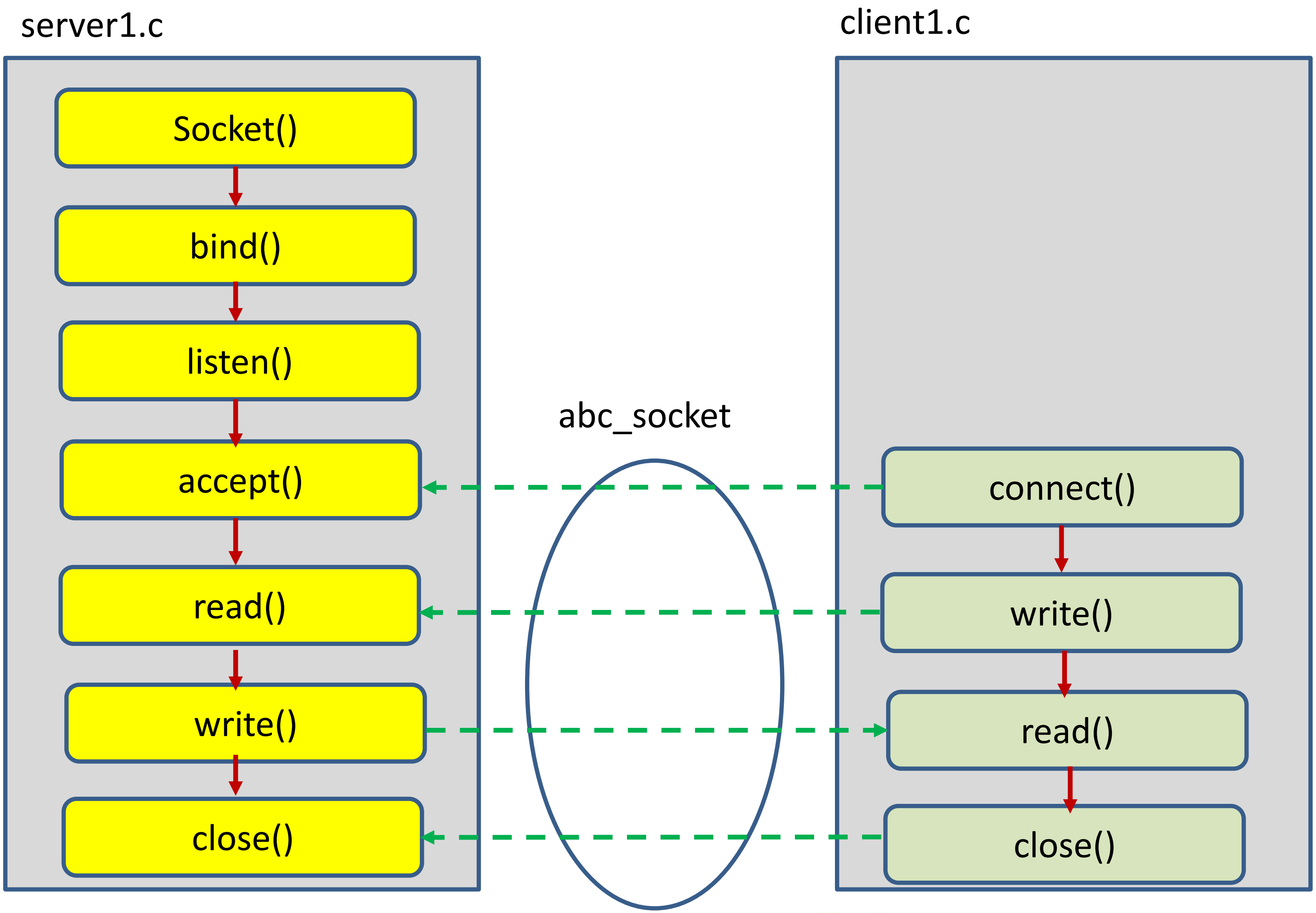
Person

Socket

communication End Point

Named Socket

Named Socket



Named socket

Domain Name is **AF_UNIX (or) AF_LOCAL**

```
struct sockaddr_un
{
    sa_family_t sun_family;      /* AF_UNIX */
    char        sun_path[108];   /* pathname */
};
```

Socket

communication End Point

unnamed Socket (Port No)

unnamed socket

Domain Name is **AF_INET**

```
struct sockaddr_in
{
   short            sin_family;   // e.g. AF_INET, AF_INET6
   unsigned short   sin_port;     // e.g. htons(3490)
   struct in_addr   sin_addr;     // see struct in_addr, below
   char             sin_zero[8];  // zero this if you want to
};
```